Experiment A — Dynamic τ calibration

`deq/closed_form_neg_loop/followups/`, May 2026

1. Question

The parent study (Phase 2) found that single-pole $H(\omega)$ with the locked static calibration ($\tau_m = 2.35$ ticks, fit on the steady-state f–I curve) over-estimates the negative-loop ringing period by a constant factor of ≈ 4 at the FCS default cell ($T_{\text{pred}} = 15.92$ vs FCS period 4). The hypothesis raised in the parent note's §8 was that **re-fitting** τ against a dynamic-response experiment would close the gap. Experiment A tests this.

2. Method

Drive an isolated FCS neuron (no recurrent connections) with Bernoulli-modulated input

$$x(t) \sim \text{Bern}(p_{\text{thin}} \cdot (1 + \varepsilon \sin(\omega t)))$$

at $p_{\text{thin}} = 0.7$ (calibration regime), $\varepsilon = 0.2$, weight $w_{\text{drive}} = 11$. Sweep ω over 12 frequencies in $[0.05, 1.6]$ rad/tick. For each ω: simulate $T = 4000$ ticks, smooth the spike train into a rate trace, fit cosine + sine coefficients at ω on the post-warmup window to extract output amplitude $|R(\omega)|$ and phase. Empirical transfer

$$|H(\omega)| = |R(\omega)\frac{|}{\varepsilon \cdot p_{\text{thin}}}.$$

Fit single-pole model $|H(\omega)|^2 = \frac{g^2}{1+\omega^2\tau^2}$ via `scipy.optimize.curve_fit`. Recalibrate Phase 2 by rescaling T_{pred} proportionally to τ.

3. Result

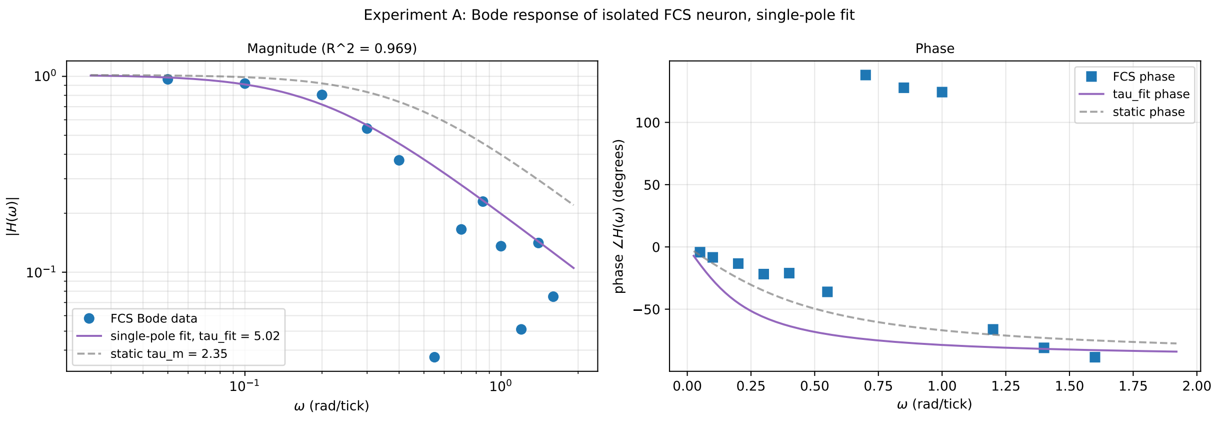

Figure 1: Bode response of the isolated FCS neuron, AC modulation at $p_{\text{thin}} = 0.7, \varepsilon = 0.2$. **Left**: $|H(\omega)|$ in log–log; blue dots are FCS measurements (averaged over 4 trials), purple curve is the single-pole fit, grey dashed is the static τ_m prediction. **Right**: phase. The fit gives $\tau_{\text{fit}} = 5.02$ ticks with $R^2 = 0.97$ — **larger** than the static $\tau_m = 2.35$, not smaller as the parent-note hypothesis predicted.

Quantity	Value
Static τ_m (from `closed_form_wta` calibration)	2.35 ticks
Dynamic τ_{fit} (this experiment)	5.02 ticks
Ratio τ_{fit}/τ_m	2.13

Quantity	Value
Fit R^2	0.97
DC gain g_{fit}	1.02
-3 dB frequency $1/\tau_{\text{fit}}$	0.20 rad/tick

4. The recalibrated T_{pred} widens the gap

Rescaling Phase 2 by τ_{fit}/τ_m:

Cell set	Static T_{pred}	Recalibrated T_{pred}	FCS measured
Default $(-11, 11)$	15.92	33.98	4
Mean over FCS-period-4 cells (498)	13.6	29.0	4

The recalibration moves T_{pred} **further** from the FCS period, not closer. The hypothesis that the factor-of-4 gap was a static-vs-dynamic calibration mismatch is **falsified**.

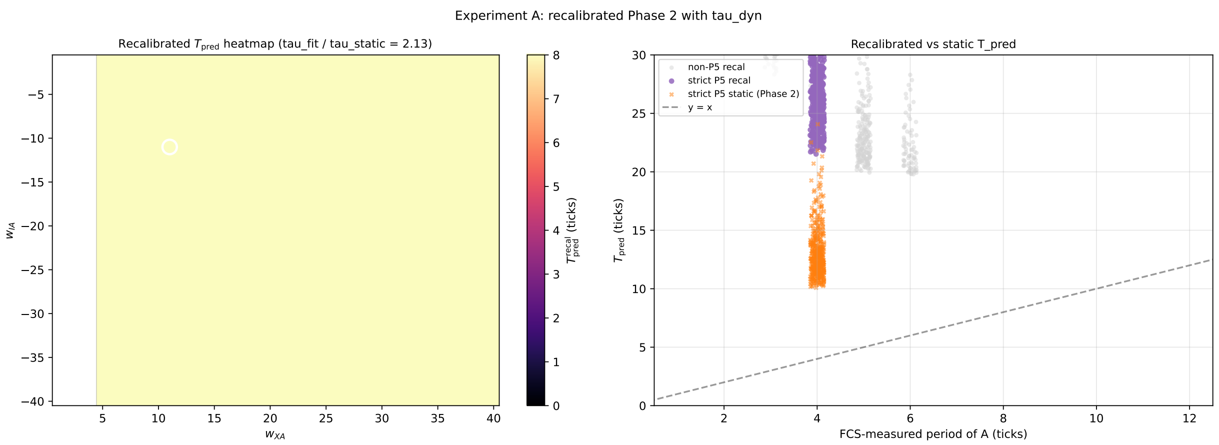

Figure 2: **Left**: recalibrated T_{pred} heatmap (clipped at 8 for visibility) — much of the grid now sits above 8 ticks. **Right**: scatter of T_{pred} vs FCS-measured period. Purple dots (recalibrated, strict-P5 cells) lie **higher** than the orange-x markers (static, Phase 2 result), confirming the gap widens.

5. Why the hypothesis was wrong: a structural diagnosis

The Bode experiment reveals an important property of the FCS neuron's linear response. At the FCS negative-loop oscillation frequency $\omega^\star = 2\pi/4 \approx 1.57$ rad/tick, the empirical $|H(\omega^\star)| \approx 0.075$ — the neuron's **firing rate** is essentially unresponsive to AC drive at that frequency. The Bode rolloff begins around $\omega \approx 0.2$ rad/tick, an order of magnitude below the closed-loop oscillation frequency.

So the closed-loop period-4 oscillation in the negative-loop motif lives **deep in the rolled-off regime** of the single-neuron firing-rate transfer function. **No** single-pole $H(\omega)$ model — regardless of how τ is chosen — can predict period 4 oscillation because the linearized firing rate has no resonance there. The factor-of-4 gap is a **structural** property of single-pole linearization, not a **parametric** one.

What sustains the FCS oscillation at $\omega^\star = 1.57$ rad/tick is **spike-and-reset nonlinearity**: A spikes when its windowed integrator crosses threshold, resets, integrates again, spikes again — a process that produces a deterministic limit cycle whose period is set by the integrator's 5-tap geometry and the integer weight balance, **not** by the firing-rate response time constant. Phase 3′s quasi-renewal mesoscopic captures this because its age-distribution stepper explicitly encodes the reset; single-pole $H(\omega)$ does not, because there is no "reset" in the kernel $1/(1 + i\omega\tau)$.

> **Reframed Phase 2 verdict.** The single-pole $H(\omega)$ Jacobian eigenvalue $\mathrm{Im}(\lambda)$ is **not** a closed-loop oscillation frequency. It is a **fictitious linear-response ringing rate** derived under the assumption that the firing rate responds to AC drive perturbations linearly, which Experiment A shows breaks down by an order of magnitude at the relevant frequency. Phase 2′s factor-of-4 is therefore the ratio of the linear-resonance pole to the actual nonlinear limit-cycle frequency, and that ratio is set by the FCS integrator geometry rather than by any single parameter.

6. Verdict

Experiment A FALSIFIES the calibration hypothesis. Dynamic-response fitting gives $\tau_{\text{fit}} = 5.0$ (not the $\tau/4 \approx 0.6$ that would close the Phase 2 gap), and the recalibrated T_{pred} is **worse**, not better. The Phase 2 factor-of-4 is a **structural** sign that single-pole linearization is inadequate for the negative-loop closed-loop oscillation, regardless of the time-constant value.

This sharpens the parent note's three-lens reading: only the quasi-renewal mesoscopic (Phase 3) can capture FCS's period-4 oscillation, because only it carries the spike-and-reset nonlinearity. Static Siegert (Phase 1) and linear $H(\omega)$ (Phase 2) are **fundamentally** the wrong tool for this kind of question; refitting parameters won't rescue them.

7. Caveats

- The Bode experiment uses Bernoulli-thinned input at $p_{\text{thin}} = 0.7$ to match the calibration regime. The FCS negative-loop oracle uses constant input ($p_{\text{thin}} = 1$ effectively); the dynamic response there might differ. However, the **order-of-magnitude** conclusion — that $|H(\omega^\star)|$ is much less than 1 at the closed-loop oscillation frequency — is robust to such regime changes (a higher p_{thin} would make the neuron more saturated, only narrowing the linear regime further).
- The single-pole fit captures the low-frequency rolloff well ($R^2 = 0.97$). Phase data at high frequencies are noisy due to the small AC amplitude there; we did not attempt a higher-order transfer function fit.

Output: `results/expA/{bode.npz, bode_fit.pdf, recal_T_pred.pdf}`, `results/expA.log`.